

# Backend Engineering Fundamentals - 6 Week Intensive Course

## Course Overview

This curriculum will take you from absolute beginner to building production-ready backend applications. By the end of 6 weeks, you'll have a solid foundation in Python backend development, RESTful APIs, databases, authentication, and deployment.

## Week 3: Databases & SQL

### Learning Objectives

- Understand relational databases and SQL
- Learn database design and normalization
- Work with PostgreSQL/SQLite
- Integrate databases with Flask using SQLAlchemy ORM
- Perform database migrations

### Resources

#### Videos:

- [SQL Tutorial - Full Course](#) by freeCodeCamp (4 hours)
- [Database Design Course](#) by freeCodeCamp (1 hour)
- [SQLAlchemy Tutorial](#) by Traversy Media (1 hour)

#### Text Resources:

- [PostgreSQL Tutorial](#) - Basics section
- [SQLAlchemy Documentation](#)
- [Real Python - SQLAlchemy Tutorial](#)

### Topics Covered

- SQL basics (SELECT, INSERT, UPDATE, DELETE)
- JOINS (INNER, LEFT, RIGHT)
- WHERE clauses and filtering
- Aggregation (COUNT, SUM, AVG, GROUP BY)
- Primary keys and foreign keys
- Database normalization (1NF, 2NF, 3NF)
- SQLAlchemy ORM models
- Relationships (One-to-Many, Many-to-Many)
- Database migrations with Alembic
- Connection pooling

### **Week 3 Project: Enhanced Book Library API with Database**

Refactor Week 2 project to use PostgreSQL/SQLite

#### **Requirements:**

- Replace in-memory storage with a database
- Create proper database schema (Books, Authors, Categories)
- Implement relationships (one author can have many books)
- Add search functionality (search books by title/author)
- Add pagination (limit and offset)
- Add filtering (filter by category, year)
- Implement database migrations
- Add data seeding script

This project tests: SQL queries, ORM usage, database design, relationships, migrations, and complex data operations.